QUANTITATIVE METHODS 1

Worksheet 3 – Dispersion and concentration

Exercises

Exercise 1 – Boxplot, potential outliers and variance choice

Consider

$$5, 6, 6, 7, 7, 8, 8, 9, 9, 10, 11, 20.$$

Use the median-of-halves convention for quartiles.

1. Calculate the five-number summary.
2. Calculate the range and IQR.
3. Calculate the fences $Q_1 - 1.5IQR$ and $Q_3 + 1.5IQR$ and identify potential outliers.
4. Calculate the mean and mean absolute deviation.
5. Calculate population variance and standard deviation if the 12 observations are the entire population.
6. Calculate sample variance and standard deviation if they are a sample.

Exercise 2 – Dispersion of grouped data

Using the 210-customer distribution from Worksheet 2, calculate $\sum n_i c_i$, $\sum n_i c_i^2$, mean, population variance, standard deviation, coefficient of variation and IQR. Compare the information provided by standard deviation and IQR.

Exercise 3 – Lorenz curve and Gini coefficient

Income values are $10, 12, 18, 25, 35$. Calculate cumulative population and income shares, construct the Lorenz curve, calculate the area by trapezoids, derive the Gini coefficient and explain why Gini measures concentration without explaining its causes.